Math-M-Addicts
MINDS THAT COUNT

SP250 Homework 6

1. There are 5 boys and 5 girls in a chess club.

   a) How many ways are there to pick 5 students and send them to a competition?
   b) How many ways are there to do so if there must be at least one boy and one girl on the team?
   c) How many ways are there to do so if there must be at least two boys and at least two girls on the team?
   d) Anya, Ben, Cecily, David and Ellie were picked. How many ways are there to assign them to play boards 1,2,3,4,5 (each student must play exactly at one board)
   e) How many ways are there to do so if Ben and David can't be sitting next to each other? (in other words, they can't play boards 1 and 2, or 2 and 3, etc.).

Solution:

---

2. How many 3-digit integers are odd and consist of 3 distinct digits?

Solution:

---

3. 101 integers are chosen from the set {1,2,..., 300}. Show that the difference between two of them is at most 2.

Solution:

---

4. Is it possible to put 8 rooks on the chessboard, so that no two rooks attack each other and exactly 3 rooks stand on white squares?

Solution:

---

5. Prove that the following equation doesn't have integer solutions: $x^2 = 21y + 10$.

Solution:

---

6. Prove that the product of any 100 consecutive positive integers is a multiple of 100! (Hint: let the largest number be n, write out the ratio that you are being asked to show is an integer and stare at it).

Solution:
